Architecture & Workflow Document

Mobile_Screen_Damage_Detection_and_Calculation_Project

This project is to solve a big e-commerce problem between sellers and buyers who are into the refurbished mobile market. When we want to exchange our phone, e-commerce companies need an estimation of our old phones as to what the condition of the phone is to determine the exact value of the phone. This project will solve this problem in a few minutes. We just need to upload the picture of the phone, and it will tell you the condition of the phone in a few seconds. It detects the damage on the screen and calculates the damage percentage for the same. The idea for this project hit my mind when I was visiting one of such refurbished websites. This project will help the industry as well as the individuals to get the exact condition of the phone.

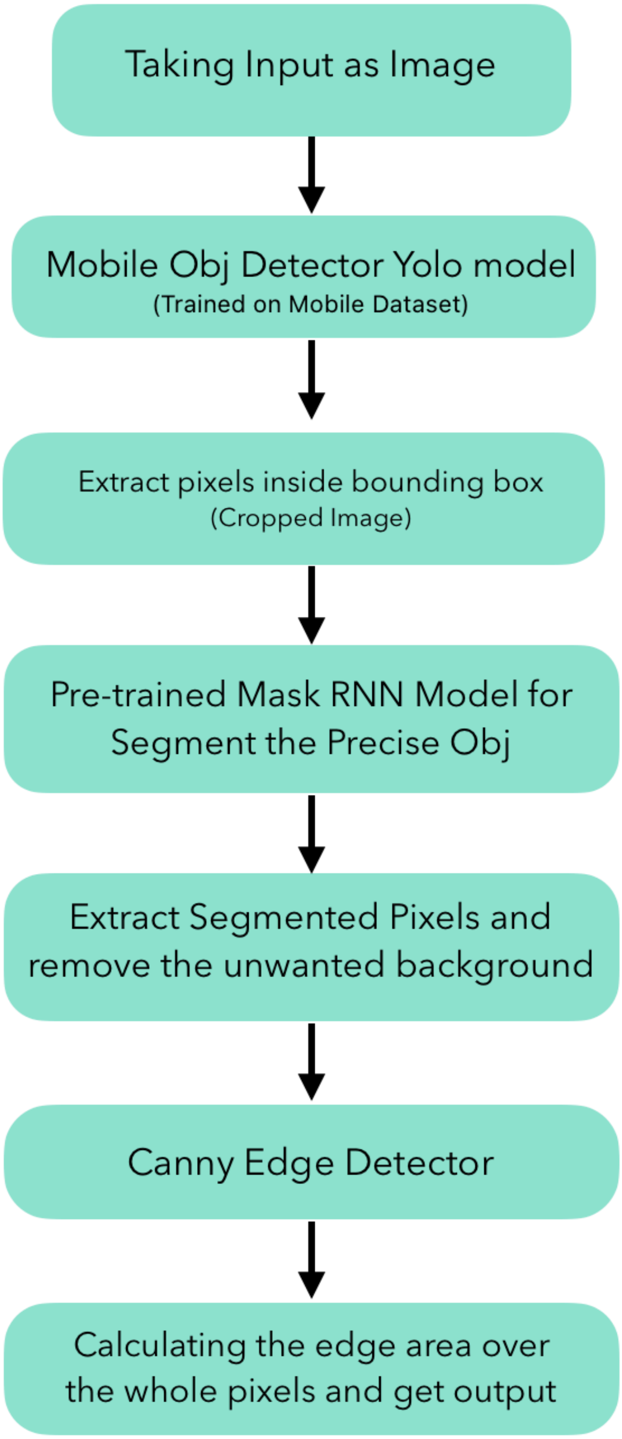
The dynamicity and some of the aspects of it may improve if the resource requirement is fulfilled. I am going to discuss all the phenomena and aspects of this project one by one with the proofs and show where we need to improve as well.

Whole code for this project is written in Python language, the dataset for the object crack screen detection is brought from the kaggle. Where I got 300 mobile phone images and their annotations with them. So this is a supervised deep learning project where I used two deep learning models , one is for object detection and another is for image segmentation (why we need these shows later in this document) and also using the edge detection technique as a canny edge detector.A special thanks to the Data Cluster Community in Kaggle, from where i get the dataset.

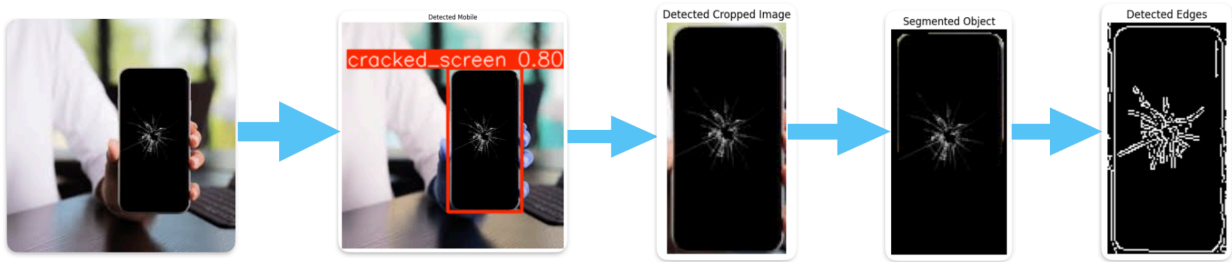
All the code, dataset and trained model is available in my git hub repository which can be easily accessible by this link

Git Hub Link for the Code

Architecture of the Project



Visualisation of how actually things are happening

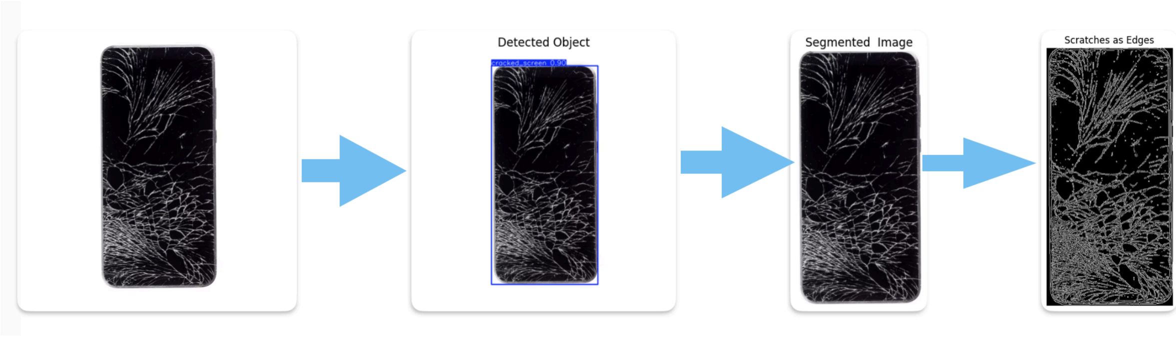


- ## Need for the Object Detection model

The need of a mobile detector does not signify the core of the project as I want it to be more dynamic, assuming that the picture of the mobile phone may carry the various background situations or may involve other objects than mobile. So to make the project more versatile, the mobile detection model will be an important decision. So it is easier for the segmentation model to find the main objective from the image. Further we take another instance where it seems more clear(we refer to the examples where we visualise the functioning). These are the results of the trained model on the validation dataset.

```
Model summary (fused): 72 layers, 3,005,843 parameters, 0 gradients, 8.1 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95):
                   all         54         52      0.868      0.758      0.854      0.639
```

By the object detector we find the main objective in the image for operations and extract in another file as a cropped image. However if we get the image in desired format as the image should be clear and aligned vertically in a right format, by doing this we minimize the load on the pipeline and would get the more accurate answer for the same. Now we see further the need of the segmentation model as shown below;



- Need for the segmentation model

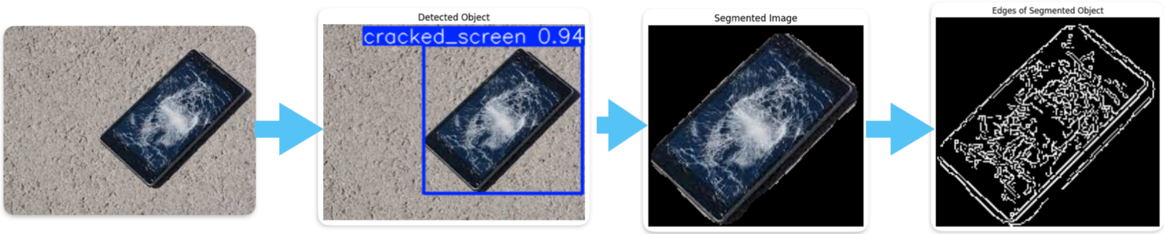
As we discussed, if we get the input image as we want(with clear background and vertically aligned) then we can skip the detector and get the output even more accurately.

The segmentation model brings us toward the more precise location for the cracks in the screen as it removes the unimportant pixels in the image and gives the more accurate plot for edge detection. By a simple illustration we understand the need for the segmentation model and the difference between the output ‘with’ and without the ‘segmentation’ model.

Results without Segmentation Model

Now we see that in edge detection there are other unwanted pixels present in the image detected by the edge detector. In such a case where the format of the input image is different from the desired format, the role of the segmentation model comes into play. Segmentation models make us precious to over need and remove the outliers from the image which lead to the desired output.Now see the results of using segmentation model ;

Results with Segmentation Model



- Some of the problems and their solutions

1. For the mobile detection model, it should not be trained on such a huge dataset due to less resources the performance of the mobile detector model can be improved.

2. And I used the pre-trained segment model as for training the model needs the segmented mobile dataset. It may lead to poor detection. However for better results we can take the input image as our desired format, it will lead us towards the good results as we see in examples.

3. In an edge detector we see some border of the mobile which will count as the edge or the damage of the mobile but in reality the case is different. We can take some margin or leverage for such borders to get the approximate result. It can be solved by good image segmentation also.